

Rahul Tammalla

Senior Data Analyst

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Professional Summary

Senior Data Analyst delivering production **SQL** analysis, **ETL** pipelines, and stakeholder-facing **Tableau/Power BI** reporting for enterprise clients, including a multi-year **GxP**-regulated engagement supporting Gilead Life Sciences. Builds **Python/PySpark** workflows on **Snowflake**, **Databricks**, and **AWS** with schema validation, row-count reconciliation, and anomaly alerting that keep **5M+** daily records audit-ready. Comfortable owning data quality, dimensional modeling, and root-cause analysis in cross-functional delivery environments.

Skills

Analytics & BI: SQL (Advanced), Tableau, Power BI, Dashboard Design, KPI Reporting, Stakeholder Reporting
Data Quality & Governance: Data Validation, Schema Enforcement, Row-Count Reconciliation, Anomaly Alerting, Dimensional Modeling, Schema Migration, Root-Cause Analysis
ETL & Data Engineering: Airflow, dbt, ETL Pipelines, PySpark, Data Cleaning
Cloud & Data Platforms: AWS (S3, Glue, Redshift, EMR, Lambda), Snowflake, Databricks, PostgreSQL, CI/CD, Kubernetes
Statistical Analysis: Hypothesis Testing, ANOVA, Chi-Square, Mann-Whitney U, k-fold CV, ROC-AUC, Multivariate Imputation
Languages & Libraries: Python, SQL, R, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn

Work Experience

BalanX Bio | Team: Multi-Modal AI Product

Senior Data Analyst

Remote
Jul 2026 – Present

- Built **Python/SQL ETL** pipelines with schema validation and quality checks across **3** signal types (voice, facial-expression, text), turning raw multi-modal input into the clean, analysis-ready dataset behind every downstream memory decision.
- Designed a sliding-window pattern-detection rule for an emotion-memory write-decision system, clustering recurring signals rather than single messages to cut false-positive writes **40%** and lift downstream retrieval relevance **20%**.
- Benchmarked **5** competing detection approaches across **4** design iterations, documenting evaluation criteria and trade-offs so the selected method could be reviewed and defended by the wider engineering team.
- Stood up a **Databricks** analytics layer with **PySpark** jobs aggregating conversation logs, emotion scores, and retrieval results by user and session, then benchmarked semantic-only against hybrid retrieval to identify which strategy surfaced more relevant memories.
- Shipped to an **AWS**-hosted production environment through **CI/CD** pipelines, running async processing on a background worker pool with automatic failover across **4** models to keep the service responsive under load.

Clarkson University | Team: Emergency Response Analytics | Client: St. Lawrence County EMS

Senior Data Analyst Assistant

Potsdam, NY
Jan 2025 – Jun 2026

- Analyzed **20M+** NEMSIS emergency records with **hypothesis testing**, **Chi-Square**, **ANOVA**, and **Mann-Whitney U** tests, engineering **70+** features to surface 15 predictors of stroke-related 911 delays, projected to cut response times by **40%**.
- Built **Python/SQL/PySpark** pipelines resolving **1M+** missing values via multivariate imputation to lift completeness to **95%**, feeding **scikit-learn**-based statistical models that raised prediction accuracy **20%** overall for the stroke-delay analysis.
- Validated models via **k-fold CV**, **ROC-AUC**, and confusion-matrix analysis, surfacing the 15 key predictors through **PostgreSQL/Tableau** dashboards as a stakeholder-facing dispatch risk-prioritization view.
- Delivered feature development and **QA testing** for **ERDPlus**, a web-based database-modeling tool, building ER-diagram, relational-schema, and auto-generated **SQL** script functionality used for downstream data delivery.

I4DI (Institute for Development Impact) | Team: Research & Program Analytics

Data Analyst

Washington, DC
May 2025 – Aug 2025

- Cut data preparation time by **43%** by building **Pandas/NumPy** preprocessing pipelines that delivered clean, analysis-ready datasets with consistent quality across all **KPI** dashboards and stakeholder reports.
- Automated survey-question generation from program-participant engagement data using structured prompting and **JSON-schema** validation, cutting manual survey-authoring effort **90%** for the analyst team.
- Optimized embedding pipelines in **PyTorch** with batched inference across **1K+** document chunks, cutting processing latency **42%** (5.1s → 2.9s) and lifting retrieval Precision@K **30%** for client-facing research queries.
- Built a document retrieval layer with **FAISS** semantic search and a real-time knowledge-base override, returning sub-second answers at **95%+** accuracy for program managers and field teams.

Cognizant | Team: Enterprise Data Analytics | Client: Gilead Life Sciences

Senior Data Analyst

Remote
Oct 2021 – Aug 2024

- Delivered **5+** **Tableau/Power BI** dashboards integrating **3+** source systems for finance and operations teams on a **GxP**-regulated enterprise analytics platform, surfacing real-time **KPIs** and cutting manual reporting effort **30%**.
- Lifted pipeline reliability **70%** and query performance **30%** by building and maintaining **Airflow/SQL ETL** pipelines on **Snowflake/Redshift**, **Databricks**, and **AWS** (S3, Glue, EMR, Lambda), keeping **5M+** daily records analysis-ready.
- Enforced data quality SLAs through automated validation checks, **row-count reconciliation**, and **anomaly alerting**, catching upstream breaks before they reached business reporting consumers.

- Designed **dimensional data models** and schema migration plans in **dbt** that standardized cross-team data delivery and kept downstream reporting stable through source-system changes.
- Ran **root-cause analysis** on recurring pipeline failures across **1M+** daily transactions, resolving defects that had been delaying **3** downstream reporting teams.
- Maintained **CI/CD** deployment pipelines on **AWS/Kubernetes** sustaining **95%+** uptime for production reporting services used by finance and operations stakeholders.

Projects

Data Analysis – NYC Public Safety Behavioral System | [Python, PySpark, Databricks, SQL, FastAPI, scikit-learn, Folium]

- Ingested and cleaned **25GB+** of NYC 311 and NYPD incident records with **PySpark** on **Databricks**, applying schema validation and data quality checks to produce an analysis-ready dataset.
- Engineered time-of-day, borough, and seasonality features, then surfaced **12+** repeatable spatial concentration patterns for non-technical city planning stakeholders.
- Trained and validated a **scikit-learn** incident-volume forecaster reaching **82%+** accuracy, benchmarked against a seasonal baseline to confirm real predictive lift.
- Served predictions through a **FastAPI** endpoint backed by a **Folium** choropleth interface, turning the analysis into a self-service tool rather than a one-off report.

Statistical Modeling – Parkinson’s Disease Screening | [Python, PCA, SMOTE, SVM, Random Forest, MLP]

- Diagnosed severe class imbalance in a clinical screening dataset, applying **SMOTE** oversampling and **PCA** dimensionality reduction to build a defensible modeling set rather than optimizing against a skewed baseline.
- Trained and compared **SVM**, **Random Forest**, and **MLP** classifiers, documenting the trade-offs so model selection could be reviewed rather than asserted.
- Validated the selected model with **k-fold cross-validation**, **ROC-AUC**, and confusion-matrix analysis, reaching **95%+** accuracy with the evaluation methodology written up alongside the result.

Interactive BI – US Accident Insights | [R, R Shiny, ggplot2, dplyr, lubridate]

- Cleaned and normalized **40,000+** US accident records with **dplyr** and **lubridate**, resolving inconsistent timestamps and location fields before any reporting layer was built.
- Built temporal and geospatial aggregations surfacing accident trends, severity distribution, and geographic risk concentration across the full record set.
- Shipped a deployed **R Shiny** application with **ggplot2** dashboards and dynamic filtering, letting non-technical users explore the data directly instead of requesting ad-hoc cuts.

NLP – Clinical Consent NER & Risk Classifier | [Python, spaCy, Transformers, scikit-learn]

- Trained a **spaCy** NER model on annotated clinical consent documents, extracting **5** entity types including treatment terms, risk disclosures, and patient obligations at **85% F1**.
- Layered a risk classifier over the NER output that flagged high-risk consent segments at **80%+** confidence, converting unstructured text into structured quality signals for downstream review.
- Applied the same document-validation pattern used in regulated reporting work: extract, validate against a defined schema, then surface exceptions rather than silently passing them through.

Education

Clarkson University <i>M.S., Applied Data Science, GPA: 3.8/4.0</i>	Potsdam, NY <i>Aug 2024 – May 2026</i>
Amrita Vishwa Vidyapeetham <i>Bachelor’s in Computer Science and Engineering</i>	Kerala, IND <i>May 2017 – Jun 2021</i>

Research

Modeling Human Preference Signals in LLM Pairwise Evaluations, Clarkson University, 2026. Analyzed **98K** crowdsourced evaluation records with statistical feature selection, identifying response verbosity as the dominant predictor of human preference and surfacing systematic bias in benchmark scoring.

Certifications & Achievements

Certificate of Completion: Claude 101, Anthropic | Jun 2026 • AI Fluency Framework & Foundations, Anthropic | Jun 2026